THE COMPLETION OF A DIRECT SUM OF COPIES OF A COMPLETE NOETHERIAN RING IS FLAT AND MITTAG-LEFFLER

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Allowed standard prerequisites (editorial).

- Universal injectivity of the completion map into the product, flatness of the cokernel of a universally injective map, and the Mittag-Leffler criterion for a pure submodule. More on Algebra, label more-algebra-lemma-ui-completion-direct-sum-into-product (proof line 2013); Algebra, labels algebra-lemma-ui-flat-domain and algebra-lemma-pure-submodule-ML (proof lines 2017-2018). Shows that the completion is a pure submodule of the product of copies of the ring, so that flatness and the Mittag-Leffler property descend from the product.
- Coherence and flatness of a product of copies of a Noetherian ring, and its Mittag-Leffler property. Algebra, labels algebra-proposition-characterize-coherent (proof line 2021), algebra-lemma-Noetherian-coherent (proof line 2023) and algebra-lemma-product-over-Noetherian-ring (proof line 2025). Gives that the product of copies of a Noetherian ring is flat and Mittag-Leffler, which combined with the purity statement gives the conclusion.

Source reference. Original lemma 05CY: sources/stacks-chow/tags.txt, source lines 6936–6936. Navigation pointer for \ref{algebra-lemma-Noetherian-coherent} used inside the counted span (line 2023); the argument that consumes it is retained in the span.

Source reference. Original lemma 05D0: sources/stacks-chow/tags.txt, source lines 6938–6938. Navigation pointer for \ref{algebra-lemma-product-over-Noetherian-ring} used inside the counted span (line 2026); the argument that consumes it is retained in the span.

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Source reference. Original lemma 059N: sources/stacks-chow/tags.txt, source lines 6821–6821. Navigation pointer for \ref{algebra-lemma-pure-submodule-ML} used inside the counted span (line 2018); the argument that consumes it is retained in the span.

Source reference. Original lemma 058P: sources/stacks-chow/tags.txt, source lines 6787–6787. Navigation pointer for \ref{algebra-lemma-ui-flat-domain} used inside the counted span (line 2017); the argument that consumes it is retained in the span.

Source reference. Original proposition 05CZ: sources/stacks-chow/tags.txt, source lines 6937–6937. Navigation pointer for \ref{algebra-proposition-characterize-coherent} used inside the counted span (line 2021); the argument that consumes it is retained in the span.

Source reference. Original lemma 05BC: sources/stacks-chow/tags.txt, source lines 6880–6880. Navigation pointer for \ref{more-algebra-lemma-ui-completion-direct-sum-into-product} used inside the counted span (line 2013); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let A be a set. Assume R is Noetherian and complete with respect to I. The completion $(\bigoplus_{\alpha \in A} R)^\wedge$ is flat and Mittag-Leffler.*

Proof. By More on Algebra, Lemma 05BC the map $(\bigoplus_{\alpha \in A} R)^\wedge \to \prod_{\alpha \in A} R$ is universally injective. Thus, by Algebra, Lemmas 058P and 059N it suffices to show that $\prod_{\alpha \in A} R$ is flat and Mittag-Leffler. By Algebra, Proposition 05CZ (and Algebra, Lemma 05CY) we see that $\prod_{\alpha \in A} R$ is flat. Thus we conclude because a product of copies of R is Mittag-Leffler, see Algebra, Lemma 05D0. □